

Chapter 9

Controlling Services and Daemons

The **systemd** daemon manages startup for Linux, including service startup and service management in general. It activates system resources, server daemons, and other processes both at boot time and on a running system.

systemd uses units to manage different types of objects.

- Service units have a **.service** extension and **represent system services**. This type of unit is used to start frequently accessed daemons, such as a web server.
- Socket units have a **.socket** extension and represent **inter-process communication (IPC)** sockets that systemd should monitor. If a client connects to the socket, systemd will start a daemon and pass the connection to it.

```
[root@localhost ~]# systemctl list-units --type=service
```

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
accounts-daemon.service	loaded	active	running	Accounts Service
atd.service	loaded	active	running	Deferred execution sch
auditd.service	loaded	active	running	Security Auditing Serv
avahi-daemon.service	loaded	active	running	Avahi mDNS/DNS-SD Stac
chronyd.service	loaded	active	running	NTP client/server
colord.service	loaded	active	running	Manage, Install and Ge
crond.service	loaded	active	running	Command Scheduler
cups.service	loaded	active	running	CUPS Scheduler
dbus-broker.service	loaded	active	running	D-Bus System Message B
dracut-shutdown.service	loaded	active	exited	Restore /run/initramfs
firewalld.service	loaded	active	running	firewalld - dynamic fi
fwupd.service	loaded	active	running	Firmware update daemon

Task	Command
View detailed information about a unit state.	<code>systemctl status <i>UNIT</i></code>
Stop a service on a running system.	<code>systemctl stop <i>UNIT</i></code>
Start a service on a running system.	<code>systemctl start <i>UNIT</i></code>
Restart a service on a running system.	<code>systemctl restart <i>UNIT</i></code>
Reload the configuration file of a running service.	<code>systemctl reload <i>UNIT</i></code>
Completely disable a service from being started, both manually and at boot.	<code>systemctl mask <i>UNIT</i></code>
Make a masked service available.	<code>systemctl unmask <i>UNIT</i></code>
Configure a service to start at boot time.	<code>systemctl enable <i>UNIT</i></code>
Disable a service from starting at boot time.	<code>systemctl disable <i>UNIT</i></code>
List units required and wanted by the specified unit.	<code>systemctl list-dependencies <i>UNIT</i></code>

```
[root@localhost ~]# systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enabl)
   Active: active (running) since Mon 2025-09-08 20:26:12 IST; 4h 30min left
     Docs: man:sshd(8)
           man:sshd_config(5)
    Main PID: 1044 (sshd)
       Tasks: 1 (limit: 11968)
      Memory: 2.7M
         CPU: 25ms
    CGroup: /system.slice/sshd.service
            └─1044 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 08 20:26:12 localhost.localdomain systemd[1]: Starting OpenSSH server daemon.
Sep 08 20:26:12 localhost.localdomain sshd[1044]: Server listening on 0.0.0.0 por
Sep 08 20:26:12 localhost.localdomain sshd[1044]: Server listening on :: port 22.
Sep 08 20:26:12 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
```